User Levels

a)Primary actors (can also act as secondary actors depending on the use case)

1. User (Abstract actor)

These processes are done by all actors they just inherit it from the user.

- Register an account.
- Log into the account.
- Log out of the account.

2. Tenants (inherits from user)

Tenants are users who search for and rent properties.They can:

- Review and rate properties
- Search properties,view and request properties
- Communicate with property owners
- Make rental payments and view booking history

3. Property owner (inherits from user)

Property owners are users who list properties for rent. They can:

- Communicate with tenants
- Add,remove or edit property listings
- Recieve and manage rental requests
- View booking history

4. Administrator (inherits from user)

Administrators manage the whole system. They can:

- Monitor and control system operations
- Remove inappropriate content and manage disputes
- View reports and system statistics

5. Manager (inherits from user)

Managers oversee property operations, ensure policies are followed and coordinate between owners, tenants and administrators. They can:

- Monitor rental requests and approvals
- Generate management reports
- Handles complaints before they go to the administrator
- Approves large discounts or promotions

6. **Financiers (inherits from user)**

Financiers handle payments, transactions and financial reporting. They can:

- Monitors payment transactions and security
- Approve refunds and adjustments that can be made
- Communicates payment issues with tenants or owners
- Generate financial reports

7. **Human Resources (HR) (inherits from user)**

HR handles user support, communication, and dispute management within the system. Responsibilities include:

- Managing complaints and disputes
- Sending notifications and reminders
- Monitoring user satisfaction and feedback
- Maintaining records of issues and resolutions

b)External Actors

Payment System
- Processes rental payments and transactions

Notification System
- Sends notifications and alerts to users regarding bookings, confirmations, or updates

List of functional requirements

Tenants

1. Tenant updates personal information
2. Tenant searches properties
3. Tenant filters properties
4. Tenant views property details and booking history
5. Tenant sends rental requests and message to property owners
6. Tenant cancels rental request
7. Tenant receives messages from property owners
8. Tenant submits rental payments
9. Tenant saves favorite properties for later viewing
10. Tenant reviews and Rate Property

Property owners

11. Property Owner updates Profile Information and Availability of Rental Properties
12. Property Owner adds Property Listing
13. Property Owner uploads Property Images
14. Property Owner edits property listing
15. Property Owner deletes property listing
16. Property Owner views rental requests

17. Property Owner accepts or reject rental request
18. Property Owner sends message to tenant
19. Property Owner views booking history
20. Property Owner marks property as unavailable or rented

Adminstrator

21. Adminstrator edits user accounts
22. Adminstrator delete user accounts
23. Adminstrator monitors rental transactions
24. Adminstrator views system reports and property listings
25. Adminstrator removes inappropriate listings
26. Adminstrator manages disputes and system settings
27. Adminstrator restores user account
28. Adminstrator suspends user account
29. Adminstrator verifies and Approve Property Listings

Managers

30. Manager monitors rental requests.
31. Manager generates management reports
32. Manager approves large discounts and promotions
33. Manager reviews complaints and forward unresolved issues

Financiers

34. Financier views transaction history
35. Financier monitors payment process and security
36. Financier communicates payment issues with tenants and owners
37. Financier approves refunds for cancelled rental transactions

HR

38. HR manages complaints
39. HR sends notifications and reminders
40. HR maintains record of user issues and resolutions

User (these 2 processes are the same for all 6 actors.They inherit from user.This was done to prevent having 6 different register, log in and log out for every actor)

41. User registers Account
42. User logs In
43. User logs Out

Non-functional Requirements

1. **Performance Requirements (the system must ensure fast response time and efficient processing under normal and high workloads)**
   - The system should load pages within 3 seconds under normal network conditions
   - The system should support at least 500 concurrent users
   - The system should process property search queries within 2 seconds in most of requests
   - The system should process payment transactions within 5 seconds under normal conditions
   - The system should maintain stable performance during its peak usage periods.
2. **Security Requirements (the system must ensure the confidentiality and protection of all users before accessing their accounts)**
   - The system should require secure authentication for all users before accessing their accounts
   - The system should encrypt user passwords

The system should protect payment information
   - The system should restrict access based on user roles.

3. **Usability Requirements (the system must be easy to learn and simple to use for all users)**
   - The system should provide user-friendly interface
   - The system should provide clear error messages when errors occur
   - The system should support responsive design to allow the system to work properly on desktops, tablets , and mobile devices.

4. **Scalability Requirements (the system must be capable of handling future growth in user, data and system functionality without ruining the performance)**
   - The system should support future expansion in the number of users and property listings without needing a major system redesign
   - The system architecture should be designed to handle increasing workloads and user requests efficiently
   - The system should allow new features and modules to be added with minimal impact on the functionality
   - The system should support database expansion to accommodate a growing number of properties, users or transactions.

5. **Maintainability Requirements (the system must be designed and allow to easy maintenance, updates and improvements)**
   - The system should allow system updates and bug fixes to be implemented without affecting the core system operations
   - The system should provide clear technical documentation for developers and system administrators

6. <u>Reliability Requirements</u> (the system must work correctly and consistent over the time)

- The system should operate without failure at least 99.5% of the time in normal conditions
- The system should recover from errors automatically or with minimal intervention
- The system should ensure data consistency even in case of unexpected shutdowns.

7. <u>Availability Requirements</u> (the system should be accessible to users when they need it)

- The system should be available all the time for all users
- The system should notify users in advance of any schedule changes
- The system should support geographically distributed access if users are from different locations.

8. <u>External Requirements</u> (regulatory requirements)

- The system shall comply with data protection regulations such as GDPR for collecting, storing, and processing personal data.
- The system shall comply with secure payment standards and regulations for processing online rental payments.
- The system shall comply with local real estate and rental laws regarding property listings and rental transactions.
- The system shall protect user privacy and ensure personal data is accessed only by authorized parties.
- The system shall generate payment records or receipts in accordance with applicable financial and tax regulation